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Assignments :-

Module – Cyber Security - Virtualization and Cloud Basics

1. What is cloud computing?

Ans. Cloud computing is the delivery of computing services – such as servers, storage, databases, networking, software, and analytics – over the internet instead of running everything on your own physical computer or local data center.

Instead of buying and maintaining hardware yourself, you rent resources from cloud providers and use them when needed.

Common Cloud providers include :

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud
- Oracle Cloud Provides

2. Describe cloud computing deploy model?

Ans. Cloud computing deployment models describe where the cloud infrastructure is hosted, who owns it, and who can access it.

The Four main deployment models are:

1. Public Cloud :-

A public cloud is owned and operated by a third-party cloud provider, and services are delivered over the internet to multiple customers.

Features :

- Shared infrastructure
- Pay-as-you-go pricing
- Highly scalable
- Managed by provider

Advantages :

- Low upfront cost
- Easy scalability
- No maintenance responsibility

Disadvantages :

- Less control
- Possible security/compliance concerns

Examples :

- Amazon Web Services
- Microsoft Azure
- Google Cloud

Use Cases :

- Website hosting
- Mobile applications
- Development and testing

2. Private Cloud :-

A private cloud is dedicated to a single organization. It may be hosted internally or by a third party.

Features :

- Exclusive infrastructure
- Greater customization
- Enhanced security and control

Advantages :

- Better privacy
- Improved compliance
- More control over resources

Disadvantages :

- Expensive
- Requires maintenance expertise

Use Cases :

- Banking systems
- Government organizations
- Healthcare applications

3. Hybrid Cloud :-

A hybrid cloud combines public and private clouds, allowing data and applications to move between them.

Features :

- Integration of both environments
- Flexible workload management

Advantages :

- Cost optimization
- Better flexibility
- Improved disaster recovery

Disadvantages :

- Complex management
- Integration challenges

Example :

An organization stores sensitive data in a private cloud but uses a public cloud for high-traffic applications.

Use Cases :

- Seasonal business workloads
- Backup and recovery
- Large enterprises

4. Community Cloud :-

A community cloud is shared by several organizations with similar requirements, such as security, policy, or compliance needs.

Features :

- Shared infrastructure among related organizations
- Cost distributed across members

Advantages :

- More secure than public cloud
- Cost-effective for specific communities

Disadvantages :

- Limited scalability
- Shared governance complexity

Use Cases :

- Universities
- Research institutions
- Government departments

3. What are components of cloud computing?

Ans. The main components of cloud computing are typically grouped into front-end, back-end, and the supporting infrastructure/services that connect them.

1. Front-End Platform :-

This is the part users interact with.

Includes:

- User devices (laptops, phones, tablets)

- Web browsers or mobile apps
- Cloud client software

2. Back-End Platform :-

This is the cloud provider's infrastructure that powers the service.

Includes:

- Servers
- Storage systems
- Databases
- Virtual machines
- Management software

3. Cloud Infrastructure :-

The physical and virtual resources needed to run cloud services.

Components:

- Data centers
- Networking hardware
- Virtualization technology
- Power and cooling systems

4. Cloud Services :-

Cloud computing delivers services in different models:

Infrastructure as a Service (IaaS)

Provides virtual hardware resources.

Examples:

- Virtual servers
- Storage
- Networking

Platform as a Service (PaaS)

Provides a development environment for applications.

Examples:

- Databases
- Runtime environments
- Developer tools

Software as a Service (SaaS)

Provides ready-to-use software over the internet.

Examples:

- Email services
- Online office tools
- CRM software

6. Security Mechanisms

Security protects cloud systems and user data.

Includes:

- Authentication
- Authorization
- Encryption
- Firewalls

- Backup and disaster recovery

7. Storage Systems

Cloud storage keeps data accessible and scalable.

Types:

- Object storage
- Block storage
- File storage

4. Cloud computing advantage and disadvantage Advantages of Cloud Computing?

Ans. **Advantages of Cloud Computing :-**

1. Cost Savings :-

- No need to buy expensive hardware and software
- Pay only for the resources you use
- Reduces maintenance and electricity costs

2. Scalability :-

- Resources can be increased or decreased easily
- Suitable for businesses with changing workloads

3. Accessibility :-

- Access data and applications from anywhere with internet
- Supports remote work and collaboration

4. Automatic Updates :-

- Cloud providers manage software updates and security patches
- Saves time for organizations

5. Backup and Disaster Recovery :-

- Data backup is easier and faster
- Helps recover data during system failures

6. High Speed and Performance :-

- Cloud services provide fast deployment and high-performance servers
- Applications can run efficiently

7. Reliability :-

- Data is stored across multiple servers and locations
- Reduces risk of downtime

8. Collaboration :-

- Multiple users can work on the same files simultaneously

Disadvantages of Cloud Computing :-

1. Internet Dependency :-

- Requires a stable internet connection
- Poor connectivity can interrupt access to services

2. Security and Privacy Risks :-

- Sensitive data is stored on third-party servers

- Risk of hacking or data breaches

3. Downtime :-

- Cloud services may become unavailable during outages
- Provider-side failures affect users

4. Limited Control :-

- Users have less control over infrastructure and configurations
- Dependence on cloud provider policies

5. Vendor Lock-In :-

- Difficult to switch from one cloud provider to another
- Migration may be costly and complex

6. Ongoing Costs :-

- Monthly or usage-based charges can increase over time
- Large-scale usage may become expensive

7. Performance Variability :-

- Shared resources can sometimes lead to slower performance